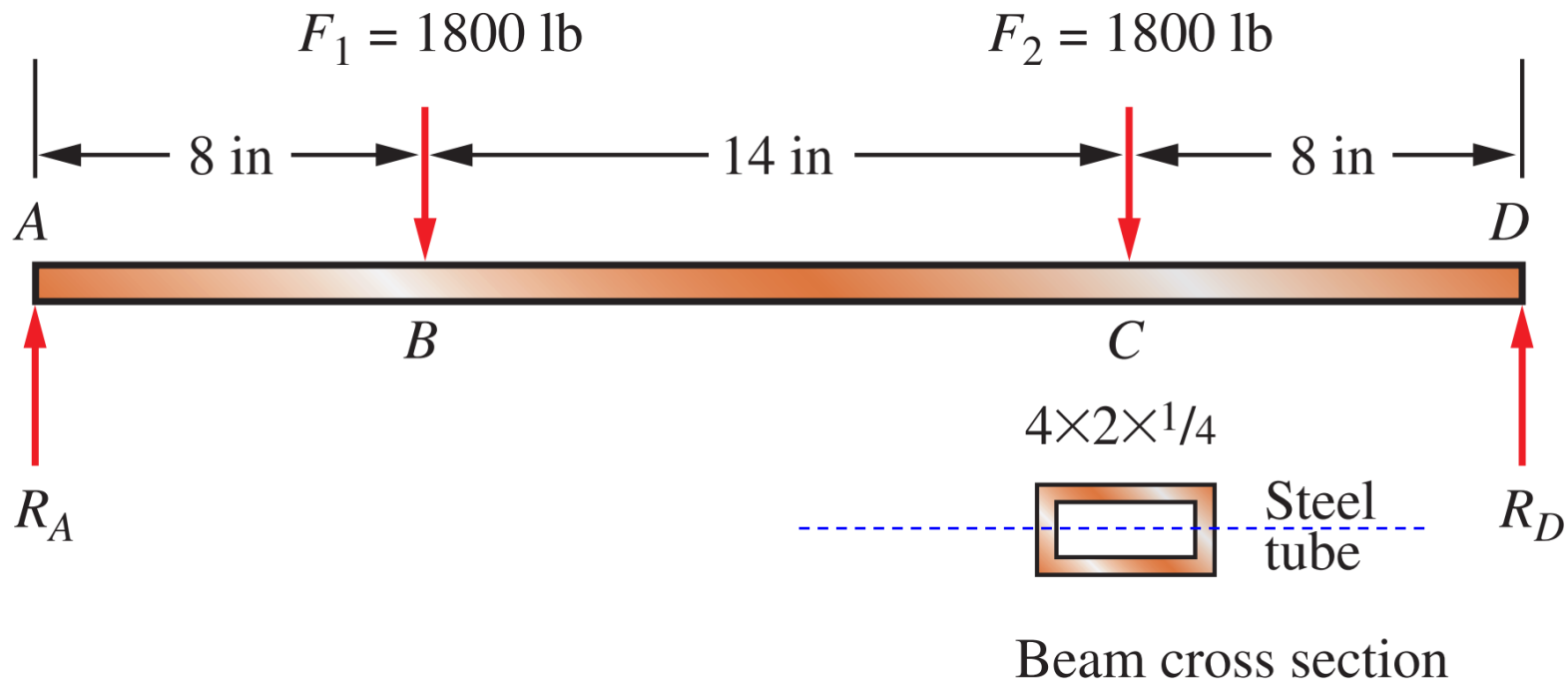


1. A part of the structure for a factory automation system is a beam that spans 30.0 in as shown in the figure. The left load $F_1 = 1800$ lb remains constantly applied, while the right load $F_2 = 1800$ lb is applied and removed frequently as the machine cycles. Determine the stress ratio for point B and C . Also, use the **maximum shear stress theory** to select an aluminum alloy (3003) from Appendix 9 for the beam with design factor of 5. The bending is along the dashed line. Use Appendix 15 to get section modulus for HSS4×2×1/4. (15' in total)



Solutions to Assignment #1:

Step 1: determine the maximum and minimum stress for point B and C .

There are two loading cases: (I) $F_2 = 1800$ lb; (II) $F_2 = 0$ lb, both with $F_1 = 1800$ lb.

Using Free-Body-Diagrams to solve the reactions at two ends and then plot the force and moment diagrams.

For point B:

The maximum bending moment for the loading case (I) and (II) are 14400 lb · in and 10560 lb · in, respectively.

With cross section bending modulus $S = 1.48$ in³ (found in [Appendix 15-14](#)), the maximum and minimum stress are

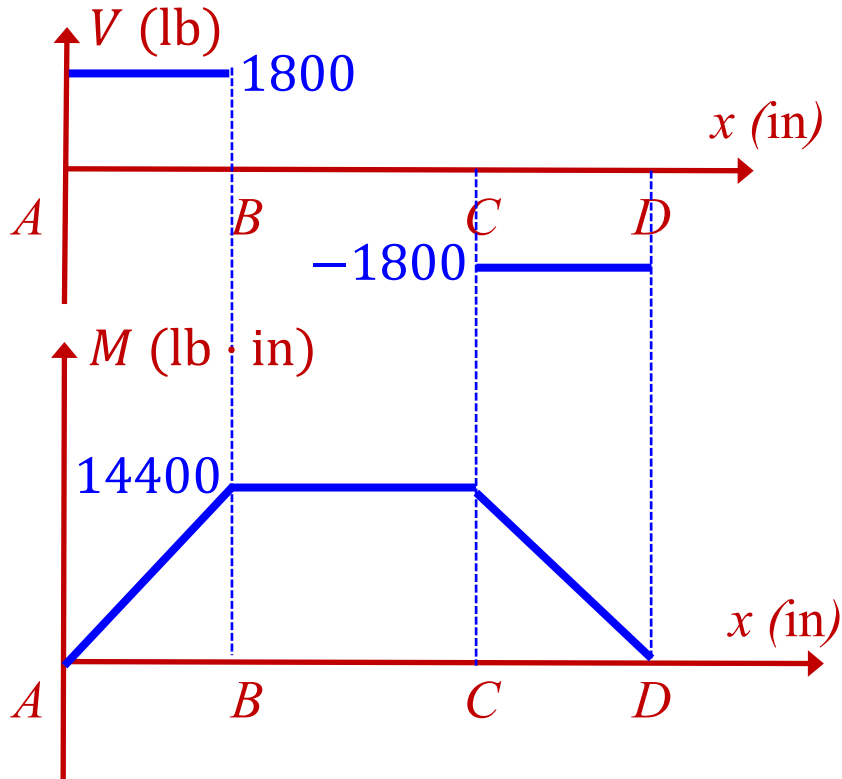
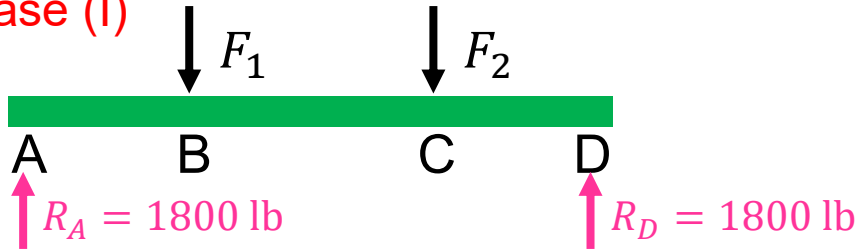
$$\sigma_{\max} = \frac{M_{\max}}{S} = \frac{14400}{1.48} = 9730 \text{ psi}$$

$$\sigma_{\min} = \frac{M_{\min}}{S} = \frac{10560}{1.48} = 7135 \text{ psi}$$

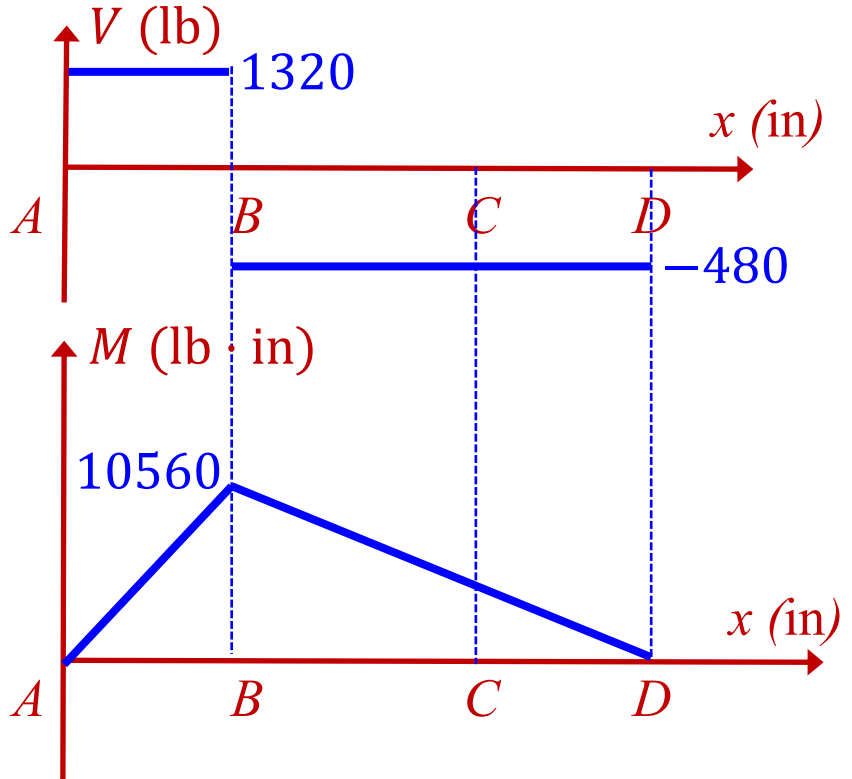
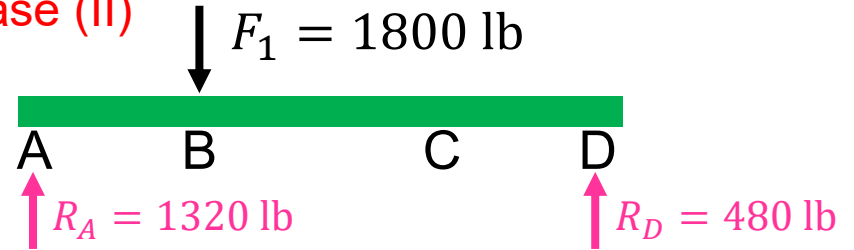
$$\text{Stress ratio } R = \frac{\sigma_{\min}}{\sigma_{\max}} = \frac{7135}{9730} = \boxed{0.733}$$

Solutions to Assignment #1, cont'd:

Case (I)



Case (II)



Solutions to Assignment #1, cont'd:

For point C:

The maximum bending moment for the loading case (I) and (II) are $14400 \text{ lb} \cdot \text{in}$ and $10560 \times \frac{8}{22} = 3840 \text{ lb} \cdot \text{in}$, respectively.

With cross section bending modulus $S = 1.48 \text{ in}^3$ (found in [Appendix 15-14](#)), the maximum and minimum stress are

$$\sigma_{\max} = \frac{M_{\max}}{S} = \frac{14400}{1.48} = 9730 \text{ psi}$$

$$\sigma_{\min} = \frac{M_{\min}}{S} = \frac{3840}{1.48} = 2595 \text{ psi}$$

$$\text{Stress ratio } R = \frac{\sigma_{\min}}{\sigma_{\max}} = \frac{2595}{9730} = \boxed{0.267}$$

Solutions to Assignment #1, cont'd:

Step 2: The case (I) induces the highest possible normal tensile stress on the surface of the beam by the maximum bending moment (14400 lb · in), which is also 2D plane stress state. The corresponding square stress state and Mohr's circle can be drawn.

With $\sigma_x = \sigma_{\max} = 9730$ psi, $\sigma_y = \tau_{xy} = 0$, we know the two principal stresses $\sigma_1 = 9730$ psi, $\sigma_2 = \sigma_3 = 0$, and the absolute maximum shear stress is

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = \frac{9730}{2} = 4865 \text{ psi}$$

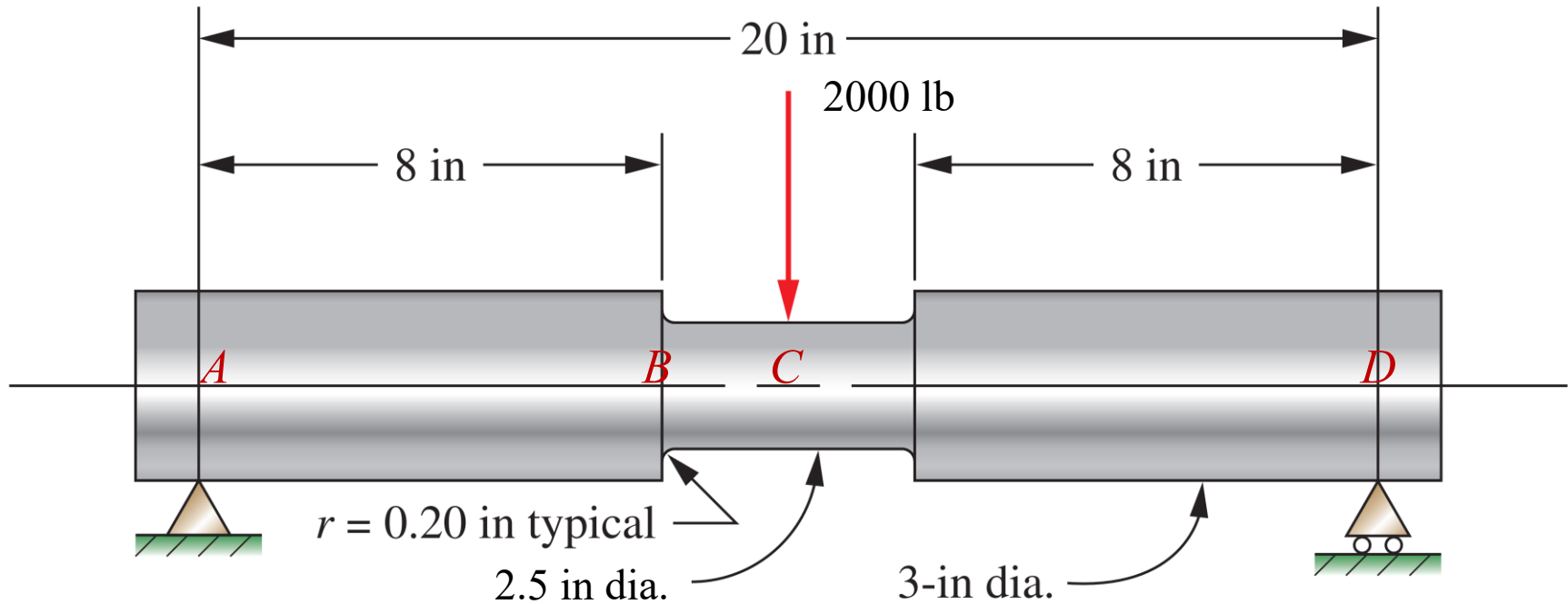
Using Maximum Shear Stress Theory (MSST) and with design factor of 5, we need to find a material with tensile yield strength

$$s_y \geq 2 \cdot N \cdot \tau_{\max}^{\text{abs}} = 2 \times 5 \times 4865 = 48650 \text{ psi} = 49 \text{ ksi}$$

To meet $s_y \geq 49$ ksi, we recommend the following aluminum alloy:

2219-T87 ($s_y = 57$ ksi) or 2024-T361 ($s_y = 57$ ksi)

2. Use Goodman criterion to determine the design factor for a beam with circular cross-section to carry a repeated central load of 2000 lb. Assume the material is wrought polished steel SAE 3140 (annealed). Take reliability as 90%. Stress concentration needs to be considered at point *B*. (15' in total)



Solutions to Assignment #2:

Step 1: Since the loading is repeated, meaning that we need to design the beam under cyclic loadings. To this end, we first need to determine the maximum and minimum stress in the beam over time.

Using Free-Body-Diagrams to solve the reactions at two ends and then plot the force and moment diagrams.

Note that, 2000 lb is the maximum loading. Thus, the mean and alternating loading (amplitude of loading) is

$$F_a = F_m = 1000 \text{ lb}$$

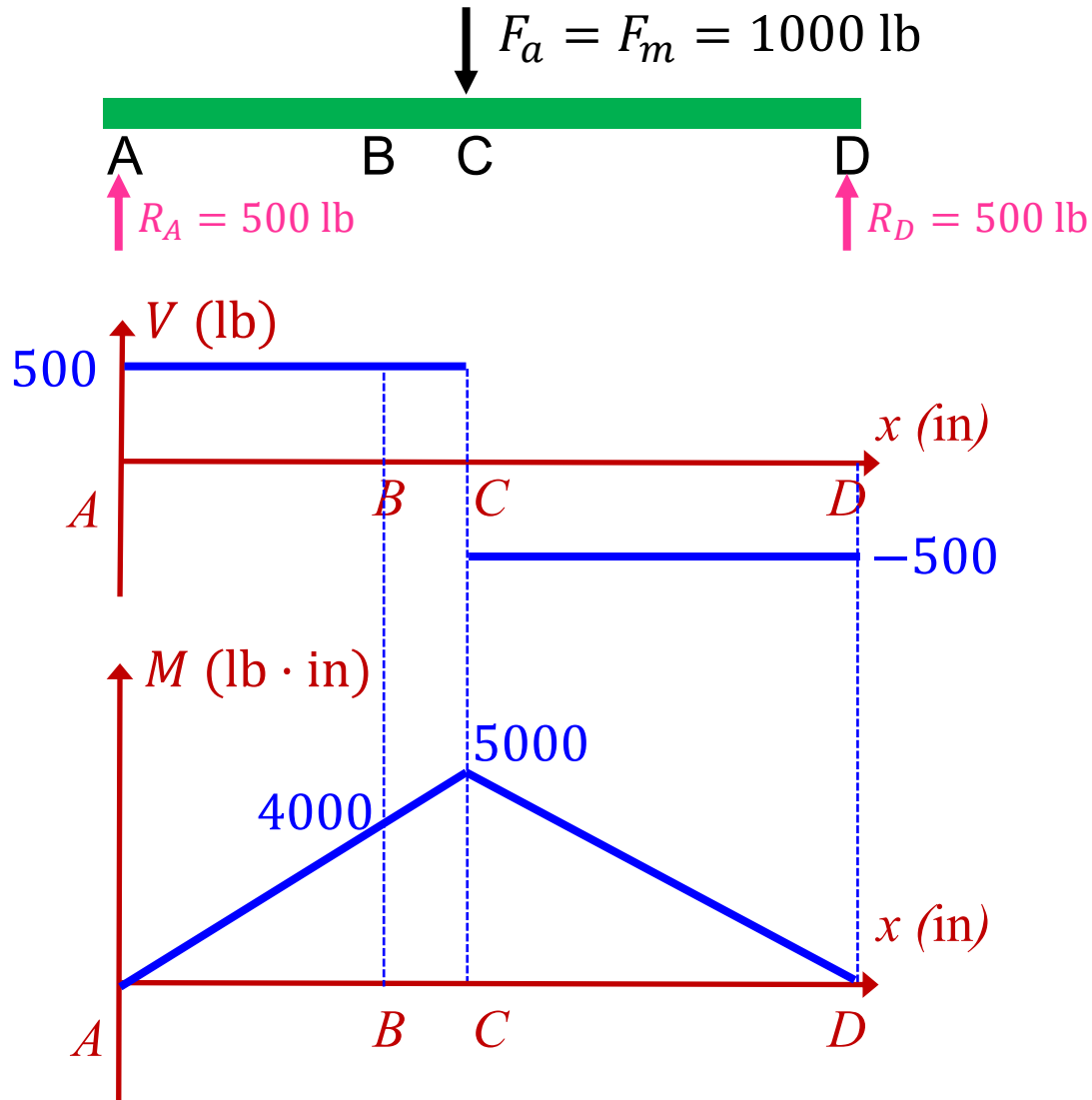
The bending moment occurs at the point B of the beam

$$M_B = 4000 \text{ lb} \cdot \text{in}$$

Then, the corresponding mean and alternating stress is

$$\sigma_m = \sigma_a = \frac{M_B}{S} = \frac{4000}{\pi \times 2.5^3 / 32} = 2608 \text{ psi}$$

Solutions to Assignment #2, cont'd:



Solutions to Assignment #2, cont'd:

Step 2: In order to use Goodman criterion, i.e.,

$$\frac{1}{N} = \frac{\sigma_m}{s_u} + \frac{K_t \sigma_a}{s_n'}$$

we need to know the estimated actual endurance limit (s_n').

The material is wrought polished steel SAE 3140 (annealed). From Appendix 3 in the textbook, we get $s_u = 95$ ksi. Then, use Figure 5-11(a) to find the corresponding endurance limit $s_n = 47.5$ ksi.

Use Eq. (5-21) in the textbook to determine s_n' :

$$s_n' = s_n \cdot C_m \cdot C_{st} \cdot C_R \cdot C_S = 47.5(1.0)(1.0)(0.90)(0.88) = 37.6 \text{ ksi}$$

Wrought steel: $C_m = 1.0$

Bending stress: $C_{st} = 1.0$

Reliability factor: $C_R = 0.90$ (Table 5-3)

Size factor: $C_S = 0.88$ (Table 5-4 using equivalent diameter 0.925 in)

(For nonrotating circular section in repeated and reversed flexure, the equivalent diameter is $D_e = 0.37D = 0.925$ in)

Solutions to Assignment #2, cont'd:

Step 3: At fillets (point B) determine stress concentration factor:

$$r/d = 0.2/2.5 = 0.08, \quad D/d = 3.0/2.5 = 1.2$$

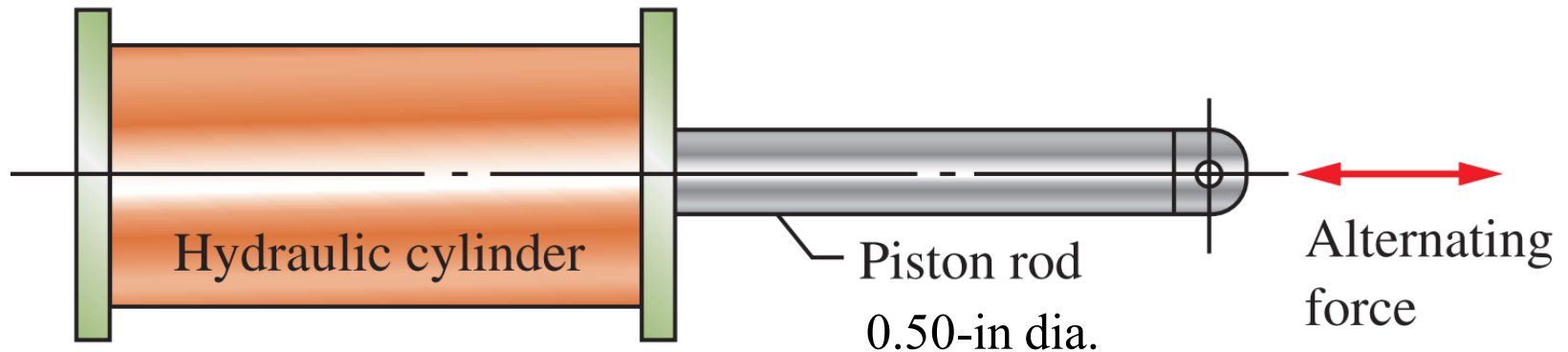
From Appendix Figure A18-6, we get the stress concentration factor $K_t = 1.76$

Finally, using Goodman criterion,

$$\frac{1}{N} = \frac{2608}{95000} + \frac{1.76 \times 2608}{37600} = 0.1495$$

$N = 6.69$

3. A hydraulic cylinder pushes a heavy tool during the outward stroke, placing a compressive load of 400 lb in the piston rod. During the return stroke, the rod pulls on the tool with a force of 1600 lb. Compute the resulting design factor for the 0.50-in-diameter rod when subjected to this pattern of forces for many cycles. The material is SAE 4130 WQT 1300 steel (cold drawn). Take reliability as 99%. (**10'** in total)



Solutions to Assignment #3:

Step 1: For cyclic loadings, we first need to determine the maximum and minimum stress in the piston rod over time.

Note that, the maximum tensile load is 1600 lb and the minimum compressive load is -400 lb. Thus, the mean and amplitude of loading is

$$F_m = \frac{1600 + (-400)}{2} = 600 \text{ lb}$$

$$F_a = 1600 - F_m = 1600 - 600 = 1000 \text{ lb}$$

Then, the corresponding mean and alternating stress is

$$\sigma_m = \frac{F_m}{A} = \frac{600}{\pi \times 0.5^2 / 4} = 3056 \text{ psi}$$

$$\sigma_a = \frac{F_a}{A} = \frac{1000}{\pi \times 0.5^2 / 4} = 5093 \text{ psi}$$

Solutions to Assignment #3, cont'd:

Step 2: The material is SAE 4130 WQT 1300 steel (cold drawn). From Appendix 3 in the textbook, we get $s_u = 98$ ksi. Then, use Figure 5-11(a) to find the corresponding endurance limit $s_n = 37$ ksi.

Use Eq. (5-21) in the textbook to determine s_n' :

$$s_n' = s_n \cdot C_m \cdot C_{st} \cdot C_R \cdot C_s = 37(1.0)(0.80)(0.81)(1.0) = 24 \text{ ksi}$$

Wrought steel: $C_m = 1.0$

Axial loading: $C_{st} = 0.80$

Reliability factor: $C_R = 0.81$ (Table 5-3)

Size factor: $C_s = 1.0$ (page 182 in the textbook)

(for any shape in repeated direct axial tensile stress, all parts of the cross-section experience the same level of tensile stress and, therefore, all parts are equally susceptible to initiation of a fatigue crack)

Solutions to Assignment #3, cont'd:

Step 3: Using Goodman criterion, i.e.,

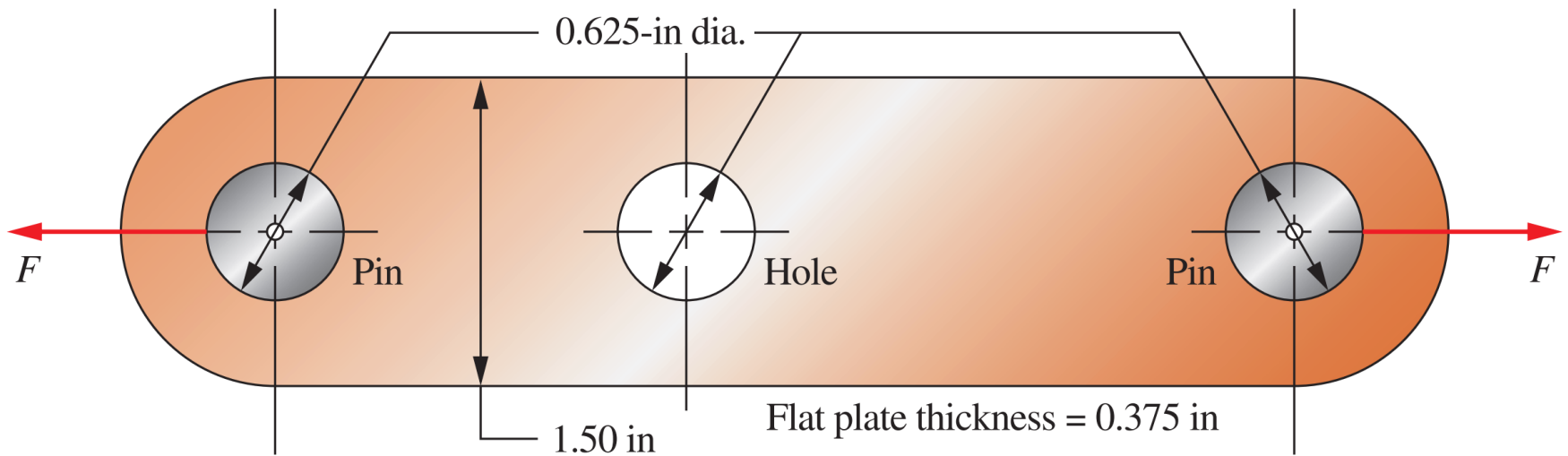
$$\frac{1}{N} = \frac{\sigma_m}{s_u} + \frac{K_t \sigma_a}{s_n'},$$

we get

$$\frac{1}{N} = \frac{3056}{98000} + \frac{1.0 \times 5093}{24000} = 0.2434$$

$N = 4.11$

4. The link is subjected to a tensile force that varies from 500 to 5500 lb. Evaluate the design factor if the link is made from SAE 1040 cold drawn steel. Take reliability as 90%. Stress concentration must be considered at both pin holes and middle hole. (**15'** in total)



Solutions to Assignment #4:

For pin holes:

For cyclic loadings, we first need to determine the maximum and minimum stress in the piston rod over time.

The maximum tensile load is 5500 lb and the minimum tensile load is 500 lb. Thus, the mean and amplitude of loading is

$$F_m = \frac{5500 + 500}{2} = 3000 \text{ lb}$$

$$F_a = 5500 - F_m = 5500 - 3000 = 2500 \text{ lb}$$

With $t = 0.375$ in, $w = 1.5$ in, $d = 0.625$ in, the corresponding mean and alternating stress is

$$\sigma_m = \frac{F_m}{(w - d)t} = \frac{3000}{(1.5 - 0.625)(0.375)} = 9143 \text{ psi}$$

$$\sigma_a = \frac{F_a}{(w - d)t} = \frac{2500}{(1.5 - 0.625)(0.375)} = 7619 \text{ psi}$$

Solutions to Assignment #4, cont'd:

Determine the stress concentration at the pin holes:

For tension load applied through a pin in the hole, we use curve B in the Figure A18-4 in the textbook to find the stress concentration, corresponding to $d/w = 0.42$,

$$K_t = 2.84$$

The material is SAE 1040 cold drawn steel. From Appendix 3 in the textbook, we get $s_u = 80$ ksi. Then, use Figure 5-11(a) to find the corresponding endurance limit $s_n = 30$ ksi.

Use Eq. (5-21) in the textbook to determine s_n' :

$$s_n' = s_n \cdot C_m \cdot C_{st} \cdot C_R \cdot C_s = 30(1.0)(0.80)(0.90)(1.0) = 21.6 \text{ ksi}$$

Wrought steel: $C_m = 1.0$

Axial loading: $C_{st} = 0.80$

Reliability factor: $C_R = 0.90$ (Table 5-3)

Size factor: $C_s = 1.0$ (page 182 in the textbook)

Solutions to Assignment #4, cont'd:

Using Goodman criterion, i.e.,

$$\frac{1}{N} = \frac{\sigma_m}{s_u} + \frac{K_t \sigma_a}{s_n'},$$

we get

$$\frac{1}{N} = \frac{9143}{80000} + \frac{2.84 \times 7619}{21600} = 1.1160$$

$N = 0.90$ (for pin holes)

Solutions to Assignment #4, cont'd:

For middle hole:

The mean and alternating stresses are the same as pin holes.

The material property is the same as before. $s_u = 80$ ksi, $s_n = 30$ ksi, and $s'_n = 21.6$ ksi.

For direct tension on plat with central hole, we use curve A in the Figure A18-4 in the textbook to find the stress concentration, corresponding to $d/w = 0.42$,

$$K_t = 2.22$$

Using Goodman criterion, we get

$$\frac{1}{N} = \frac{9143}{80000} + \frac{2.22 \times 7619}{21600} = 0.8974$$

$N = 1.11$ (for middle hole)

5. A standard 3-in schedule 40 steel pipe is proposed to be used to support the roof of a porch during renovation. Its length is 15 ft. The pipe is made from ASTM A501 structural steel. Determine the safe load on the pipe to achieve a design factor of 3 if the pipe is straight. All materials and cross-section properties can be found in Appendix in the textbook. (10' in total)

Solutions to Assignment #5:

For centrally loaded column:

Step 1:

From **Appendix Table 15-17**, for standard 3-in schedule 40 steel pipe the cross-section properties are:

Cross-section area $A = 2.228 \text{ in}^2$

Radius of gyration is $r = 1.164 \text{ in}$

Step 2:

Slenderness Ratio $S_r = \frac{KL}{r} = \frac{1.0 \times 15 \times 12}{1.164} = 154.6$

Step 3:

The material is ASTM A501 structural steel. From **Appendix 7**, we find the yield strength $s_y = 36 \text{ ksi}$ and stiffness (Young's modulus) $E = 29 \text{ Mpsi}$.

Step 4:

The transition slenderness ratio $C_c = S_{rT} = \sqrt{\frac{2\pi^2 E}{s_y}} = \sqrt{\frac{2\pi^2 (29 \times 10^6)}{36 \times 10^3}} = 126.1$.

Since $S_r > C_c$, the column is long (Euler column).

Solutions to Assignment #5, cont'd:

For centrally loaded column:

Step 5:

Using Euler formula Eq. (6-6), we get the critical load

$$P_{cr} = \frac{\pi^2 EA}{S_r^2} = \frac{\pi^2 \times 29 \times 10^6 \times 2.228}{154.6^2} = 26680 \text{ lb}$$

Step 6:

$$\text{Allowable load } P_a = \frac{P_{cr}}{N} = \frac{26680}{3} = \boxed{8893.5 \text{ lb}}$$

6. Design a suitable diameter of a solid, round cross section for a machine link if it is to carry 12000 lb of axial compressive (centrical) load. The length of the link is 25 inch, and the two ends are pinned. Use a design factor of 3 and the material is SAE 1050 hot-rolled steel. (**10'** in total)

Solutions to Assignment #6:

The column is centrally loaded.

Since the diameter of the link is unknown (to be determined), we do not know whether the column is long (Euler column) or short (Johnson column). However, we can first assume that the column is long.

For pinned-pinned shown in Figure 6-3(a), $K = 1.0$.

The material is SAE 1050 hot-rolled steel. From Appendix 3, we know that the yield strength is $s_y = 49$ ksi and modulus of elasticity $E = 30$ Mpsi.

From Equation (6-11) in the textbook, the required diameter for a long, solid circular column is:

$$D = \left[\frac{64NP_a(KL)^2}{\pi^3 E} \right]^{1/4} = \left[\frac{64(3)(12000)(1.0 \times 25)^2}{\pi^3 (30 \times 10^6)} \right]^{1/4} = 1.115 \text{ in}$$

The radius of gyration can now be determined (according to Appendix 1):

$$r = \frac{D}{4} = \frac{1.115}{4} = 0.27875 \text{ in}$$

$$\text{The corresponding Slenderness Ratio } S_r = \frac{KL}{r} = \frac{1.0 \times 25}{0.27875} = 89.7$$

Solutions to Assignment #6, cont'd:

The transition slenderness ratio $C_c = S_{rT} = \sqrt{\frac{2\pi^2 E}{s_y}} = \sqrt{\frac{2\pi^2(30 \times 10^6)}{49 \times 10^3}} = 109.9$.

Since $S_r < C_c$, the column is actually short, i.e., Johnson column. This also means that, our presumed condition is not correct, and we should use Johnson formula to redo the calculation.

From Eq. (6-12) in the textbook, the required diameter for a short, solid circular column is

$$D = \left[\frac{4NP_a}{\pi s_y} + \frac{4s_y(KL)^2}{\pi^2 E} \right]^{1/2} = \left[\frac{4(3)(12000)}{\pi(49000)} + \frac{4(49000)(1.0 \times 25)^2}{\pi^2(30 \times 10^6)} \right]^{1/2} \\ = 1.1615 \text{ in}$$

The corresponding radius of gyration $r = \frac{D}{4} = \frac{1.1615}{4} = 0.290375 \text{ in}$

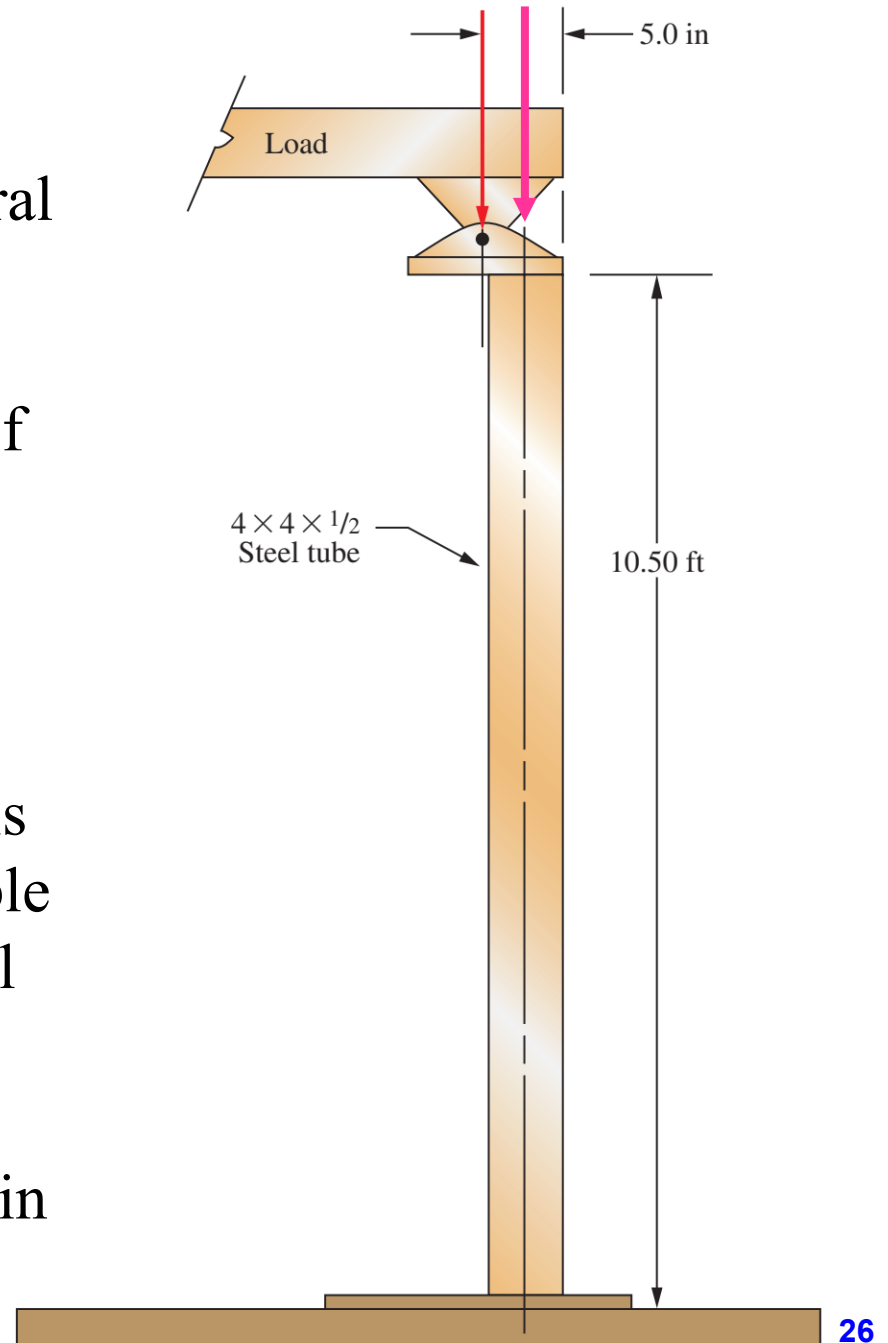
Checking Slenderness Ratio again:

$$S_r = \frac{KL}{r} = \frac{1.0 \times 25}{0.290375} = 86.1$$

This is still less than the transition slenderness ratio, so our new design is acceptable.

Finally, the designed diameter of the link is 1.1615 inch.

7. A $4 \times 4 \times \frac{1}{2}$ steel hollow tubing column is made from ASTM A500 (Shaped) Grade B structural steel. First, determine the maximum allowable central load to achieve a design factor of 3. Second, to accommodate a special mounting restriction, sometimes the load is applied eccentrically as shown. Determine whether the column is still safe when the same allowable load is eccentrically applied. All materials and cross-section properties can be found in Appendix in the textbook. (10' in total)



Solutions to Assignment #7:

For centrally loaded column:

Step 1:

From **Appendix Table 15-14**, for HSS $4 \times 4 \times \frac{1}{2}$ the cross-section properties are:

Cross-section area $A = 6.02 \text{ in}^2$

Radius of gyration is $r = 1.41 \text{ in}$

Step 2:

Slenderness Ratio $S_r = \frac{KL}{r} = \frac{1.0 \times 10.5 \times 12}{1.41} = 89.4$

Step 3:

The material is ASTM A500 (Shaped) Grade B structural steel. From **Appendix 7**, we find the yield strength $s_y = 46 \text{ ksi}$ and stiffness (Young's modulus) $E = 29 \text{ Mpsi}$.

Step 4:

The transition slenderness ratio $C_c = S_{rT} = \sqrt{\frac{2\pi^2 E}{s_y}} = \sqrt{\frac{2\pi^2 (29 \times 10^6)}{46 \times 10^3}} = 111.6$.

Since $S_r < C_c$, the column is short column (Johnson column).

Solutions to Assignment #7, cont'd:

For centrally loaded column:

Step 5:

Using Johnson formula Eq. (6-9), we get the critical **centrical** load

$$P_{cr} = A s_y \left[1 - \frac{s_y S_r^2}{4\pi^2 E} \right] = 6.02(46000) \left[1 - \frac{46000(89.4^2)}{4\pi^2 (29 \times 10^6)} \right] = 187994 \text{ lb}$$

Step 6:

The corresponding allowable load $P_a = \frac{P_{cr}}{N} = \frac{187994}{3} =$ 62665 lb

Solutions to Assignment #7, cont'd:

For **eccentrically** loaded column:

Cross-section area $A = 6.02 \text{ in}^2$

Radius of gyration is $r = 1.41 \text{ in}$

Eccentricity $e = 5 - \frac{4}{2} = 3 \text{ in}$

Distance from neutral axis to outside of section $c = \frac{B}{2} = \frac{4}{2} = 2 \text{ in}$

Then, using Eq. (6-15) we get the required yield strength

$$\begin{aligned} s_y|_{\text{req.}} &= \frac{NP_a}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{KL}{2r} \sqrt{\frac{NP_a}{AE}} \right) \right] \\ &= \frac{3(62665)}{6.02} \left[1 + \frac{(3)(2)}{1.41^2} \sec \left(\frac{1.0(126)}{2(1.41)} \sqrt{\frac{3(62665)}{6.02(29 \times 10^6)}} \right) \right] \\ &= 125505 \text{ psi} = 125.5 \text{ ksi} \end{aligned}$$

Since $s_y|_{\text{req.}} > s_y = 40 \text{ ksi}$, the column **is not safe** for the same loading that is eccentrically applied with off-distance 3 in.

8. Using the boundary-lubricated approach, design a plain surface bearing to carry a radial load of 75 lb from a shaft having a minimum acceptable diameter of 0.50 in and rotating at 600 rpm. Use an L/D ratio of the bearing to be 1.5. Compute the pV factor and specify a material from Table 16–1. (**5'** in total)

Solutions to Assignment #8:

Step 1:

Trial diameter: try $D = D_{\min} = 0.50$ in

Step 2:

$L/D = 1.5$, then the nominal length of the bearing is

$$L = 1.5D = 1.5(0.50) = 0.75 \text{ in}$$

Step 3:

The bearing pressure (using the projected area of the bearing)

$$p = \frac{F}{LD} = \frac{75}{(0.75)(0.50)} = 200 \text{ psi}$$

Step 4:

The linear speed of the journal surface

$$V = \frac{\pi Dn}{12} = \frac{\pi(0.50)(600)}{12} = 78.54 \text{ ft/min}$$

Step 5:

The pV factor

$$pV = (200)(78.54) = 15708 \text{ psi} \cdot \text{fpm}$$

$$\text{Design value of } pV = 2(15708) = 31416 \text{ psi} \cdot \text{fpm}$$

Step 6:

From Table 16-1, we can choose Porous bronze bearing ($pV = 50000 \text{ psi} \cdot \text{fpm}$).

9. Design a plain surface journal bearing to carry a steady radial load of 2.25 kN while the shaft rotates at 2200 rpm. The shaft stress analysis determines that the minimum acceptable diameter at the journal is 25 mm. Use $L/D = 2.0$ and specify the journal nominal diameter, the bearing length, the diametral clearance, the minimum film thickness of the lubricant during operation, the lubricant type that meets the requirement. Also compute the coefficient of friction, the friction torque, and the power dissipated as the result of friction. (**10'** in total)

Solutions to Assignment #9:

$F = 2.25 \text{ kN}$, $D_{\min} = 25 \text{ mm}$, $n = 2200 \text{ rpm}$, machine tool

Let $D = 25 \text{ mm}$, $R = D/2 = 12.5 \text{ mm}$

$L = 2D = 2(25) = 50 \text{ mm}$

The bearing pressure is

$$p = \frac{F}{LD} = \frac{2250}{(0.05)(0.025)} = 1.8 \text{ MPa}$$

On page 626 in the textbook, it says “General-purpose industrial machinery bearings typically operate at approximately 200 to 500 psi (1.4 to 3.4 MPa) bearing pressure”. So, the above bearing pressure is acceptable.

From Figure 16-4 in the textbook, we estimate $C_d = 40 \text{ }\mu\text{m}$. Then, $C_r = \frac{C_d}{2} = 20 \text{ }\mu\text{m}$, $\frac{R}{C_r} = \frac{12500}{20} = 625$

Solutions to Assignment #9, cont'd:

Minimum film thickness $h_o = 0.00025D = 0.00025(25) = 0.00625$ mm

$$\frac{h_o}{C_r} = \frac{0.00625}{0.02} = 0.31$$

From Figure 16-8 in the textbook ($L/D = 2$), we find the Sommerfeld Number $S = 0.08$

$$n_s = \frac{2200}{60} = 36.67 \text{ rev/s}$$

Then, the required dynamic viscosity of the lubricant is

$$\mu = \frac{SP}{n_s(R/C_r)^2} = \frac{0.08(1.8 \times 10^6)}{36.67(625)^2} = 0.01 \text{ Pa} \cdot \text{s}$$

From Figure 16-7(a), we know that, SAE 10W has $\mu = 0.01 \text{ Pa} \cdot \text{s}$ at 70°C , which meets the requirement.

Solutions to Assignment #9, cont'd:

From Figure 16-9, we find the corresponding coefficient of friction:

$$f(R/C_r) = 2.5$$

$$f = \frac{f\left(\frac{R}{C_r}\right)}{\frac{R}{C_r}} = \frac{2.5}{625} = 0.004$$

According to Eq. (16-8), the Friction Torque is

$$T_f = fFR = 0.004(2250 \text{ N})(0.0125 \text{ m}) = \boxed{0.11 \text{ N} \cdot \text{m}}$$

According to Eq. (16-9), the Frictional Power Loss is

$$P_f = \frac{T_f n}{63000} = (0.11 \text{ N} \cdot \text{m}) \left(36.67 \frac{\text{rev}}{\text{s}}\right) \left(\frac{2\pi \text{ rad}}{\text{rev}}\right) = 25.3 \frac{\text{N} \cdot \text{m}}{\text{s}} = \boxed{25.3 \text{ W}}$$